BINF 4070 / COMS 4995 W008

The Future of Personal Health Assistant

Fall 2026

HCI for Personal Health Assistants: Interface, Trust & Uncertainty

Week 08

Dr. Xuhai “Orson” Xu | Dr. Eugene Wu

AI @ VP&S Initiative

COLUMBIA | VAGELOS COLLEGE OF PHYSICIANS AND SURGEONS

COLLABORATORY AT COLUMBIA UNIVERSITY

DAP Lab

Today's Outline

1. **Understand Users and Tasks**
2. **Design and Prototype the Interaction**
3. **Implement the Interaction**
4. **Other UI/UX Design Principles**
5. **Test and Iterate**
6. **Mid-Point Presentation**

Understand Users and Tasks

What is the person trying to accomplish, and what makes it difficult?

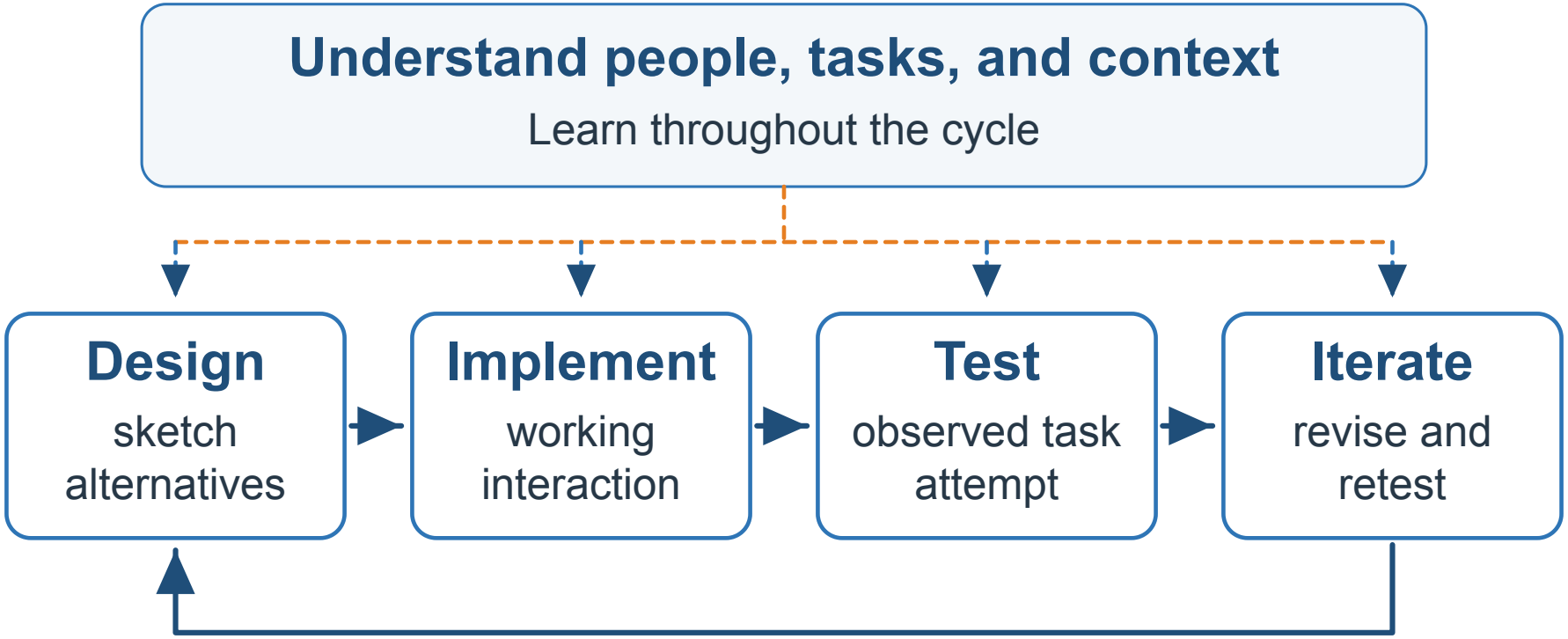
A Tired Caregiver Has a Task

Example case: Alex cares for a parent and works rotating shifts. Before leaving for work, Alex asks, “Why am I tired recently?”

Alex needs to inspect a recent sleep change and prepare questions for a clinician. Daytime sleep, interruptions, and limited time shape the interaction.

> **Human-computer interaction (HCI)** studies and designs how people use and understand computing systems in context.

User-Centered Design Is a Loop



User-centered design (UCD) grounds this loop in people's tasks and contexts.

Course diagram; human-centered design — NIST; iterative design — Design Council

Observe the Work; Ask About It

Contextual inquiry combines observation in the person's setting with questions about what they are doing.

In Alex's example session	What the designer learns
Ask: “Show me the last time you checked sleep.”	A concrete sequence, rather than a feature wish list
Observe: Alex switches between sleep data and a shift calendar.	A comparison needs schedule context
Ask: “What were you checking there?”	A “night” label may make daytime sleep hard to find

Confirm your interpretation with Alex. One session suggests a requirement to investigate; it does not describe every caregiver.

Turn the Task Into a Requirement

Task analysis identifies the actions and information a person needs to reach a goal.

Layer	Alex's example
Goal	Prepare an accurate account of a sleep change
Task	Compare recorded sleep, check gaps, add shift context
Requirement	Alex can check whether a daytime episode is included and add context
Candidate feature	A dated episode list with an edit control

Test the requirement by observing a task attempt. A request for “a dashboard” does not yet specify success.

Design and Prototype the Interaction

Make the next action understandable before making the screen attractive.

Co-design Creates Alternatives Together

Co-design means designing **with users** through collaborative creation: users and designers develop possibilities together.

Alex example: In a workshop, Alex and a designer sketch two sleep screens: one starts with a weekly comparison; one starts with a dated episode list.

They combine the comparison with a clear route to daytime episodes and shift context. This proposed design still needs testing.

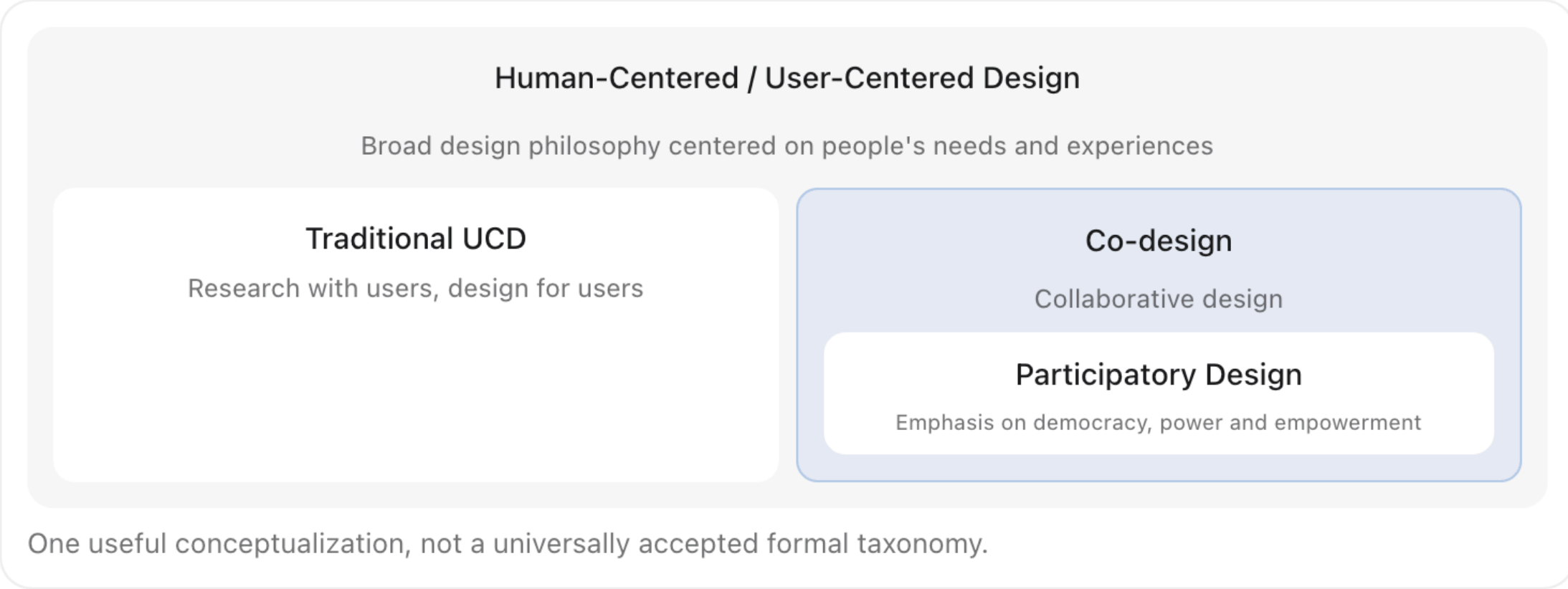
Participatory Design Shares Decision Influence

Participatory design (PD) designs **with users as partners** who help frame the problem, create alternatives, and influence decisions. Agency, empowerment, and shared decision power are central.

Alex example: Alex and caregivers are ongoing design partners who jointly approve prototype priorities.

- They challenge a score-first plan and choose easier review of daytime sleep.
- After sketching and testing alternatives together, they prioritize dated shift notes; the next prototype adds these instead of another score display.

UCD, Co-design, and Participatory Design



Traditional UCD: FOR users · Co-design: WITH users · PD: WITH users and shared decision power

Relationship diagram — supplied by Orson

Choose a Prototype for the Question

A **prototype** models an interaction for testing; **low fidelity** uses rough screens.

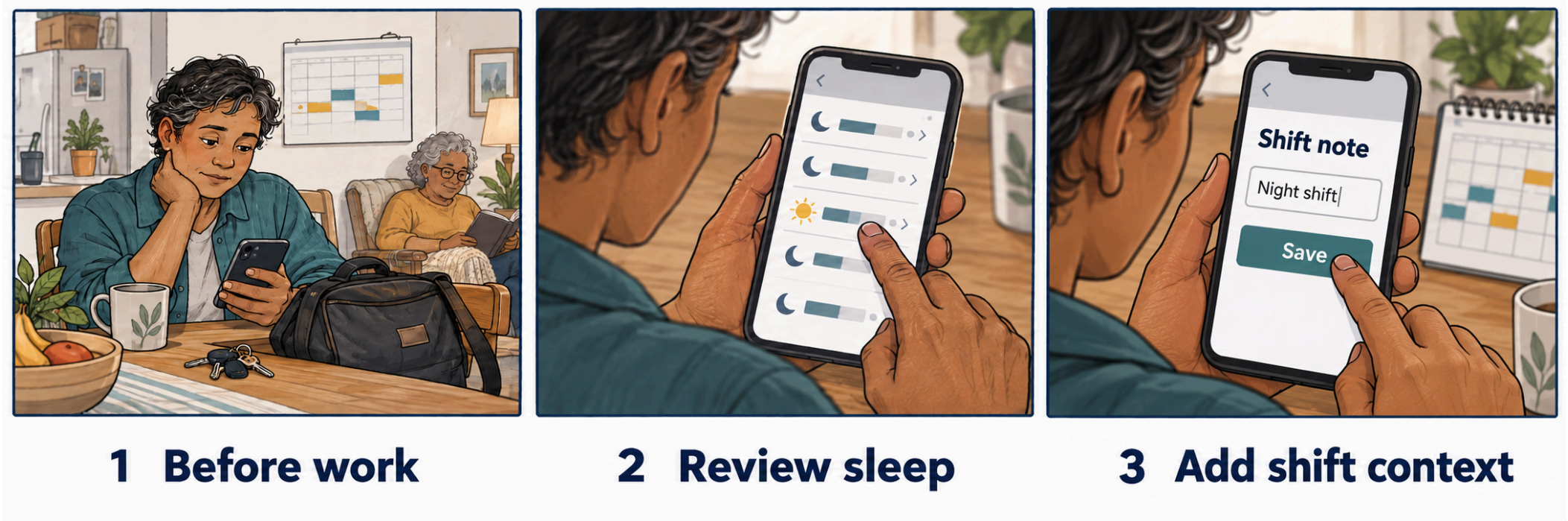


Test **labels/layout on paper**, **navigation in clickable screens**, and **saving/recovery in working software**.

Fidelity means visual and behavioral detail. More is not always better or the next required stage; a polished screen may still mock behavior.

A Storyboard Shows Use Over Time

A **storyboard** uses a sequence of illustrated scenes to show a person, their situation, and an interaction over time.



The storyboard shows **why and when** Alex needs the screen. A prototype tests **what Alex can do** with it.

Try the Alternatives With Alex

Illustrative co-design session:

Step	What Alex and the designer do
Locate the moment	Use the storyboard to focus on checking sleep before work
Try paper alternatives	Alex looks for daytime sleep on both; “nights” suggests it is excluded
Choose together	Keep the comparison with a clear route to dated **sleep episodes**

A later **clickable, higher-fidelity prototype** asks the next question: can Alex move from the comparison to an episode, add context, and return?

Reveal Detail Without Hiding the Limit

Progressive disclosure shows essential information first and opens supporting detail when requested.

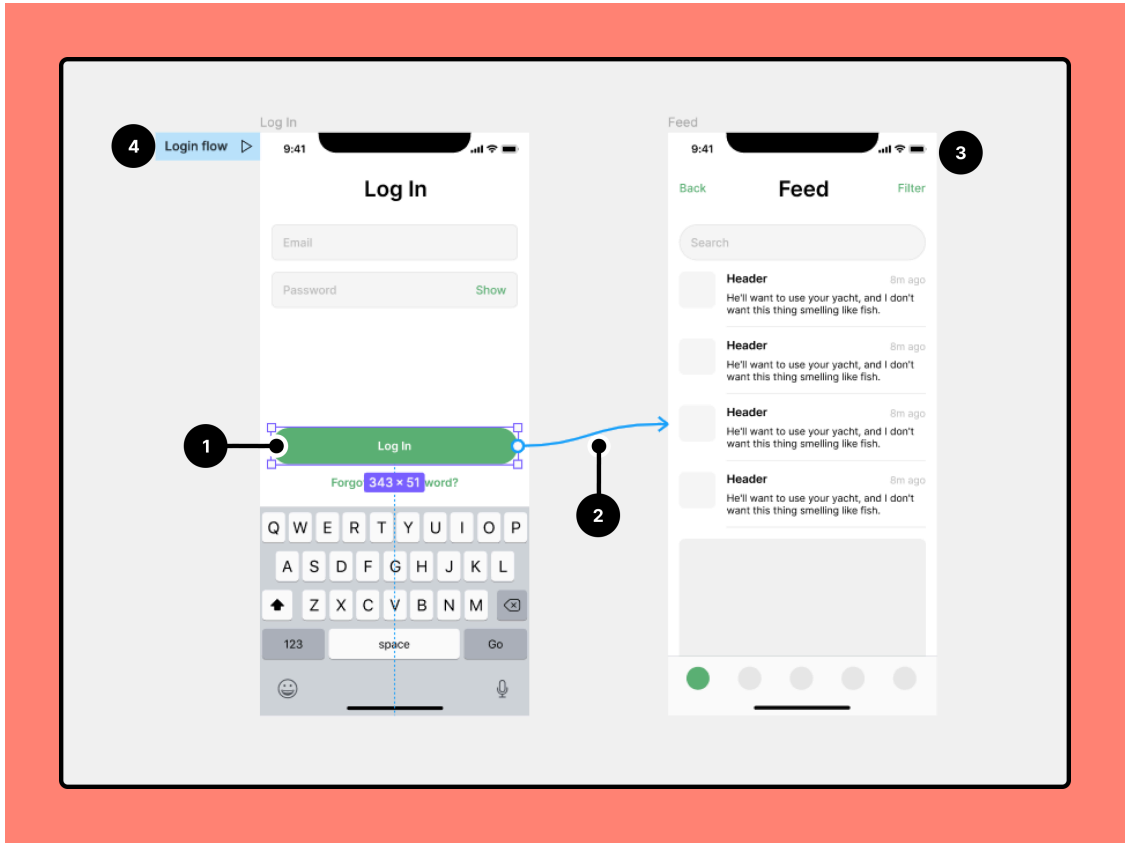
Always visible	Open on request
Source, dates, and coverage	Individual sleep episodes
Main observation and its limits	How the displayed summary was made
Available next action	An editable context form

Alex sees the sleep comparison and its coverage together, then can inspect episodes or add context. Keep a clear route back.

Figma: Draw Screens and Connect Them

- Use **frames, shapes, and text** to build a key screen.
- Optionally connect screens to test navigation.
- Keep the screen and design-process notes in one file; share a **view link**.

figma.com

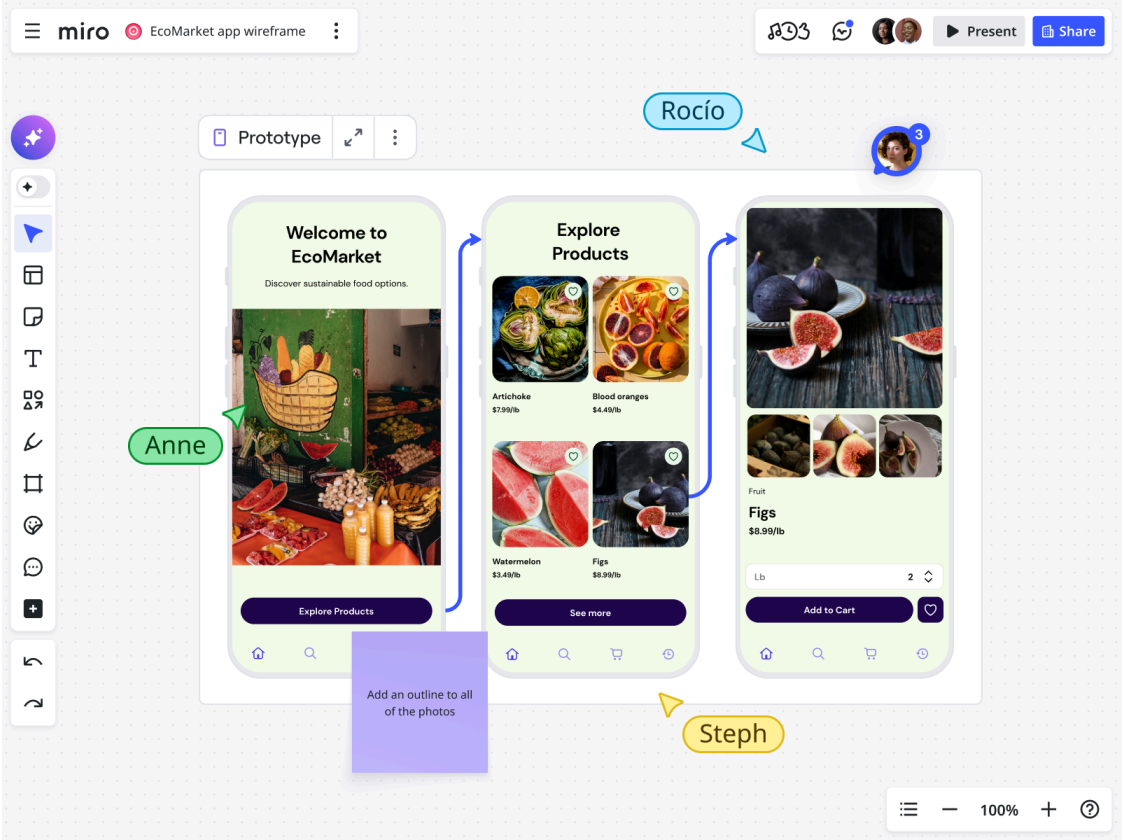


Interface screenshot — Figma, Guide to Prototyping in Figma

Miro: Sketch and Discuss on a Board

- Arrange **shapes, text, and arrows** into a screen or flow.
- Put task observations and proposed changes beside the sketch.
- Keep the work on one board; share a **view link**.

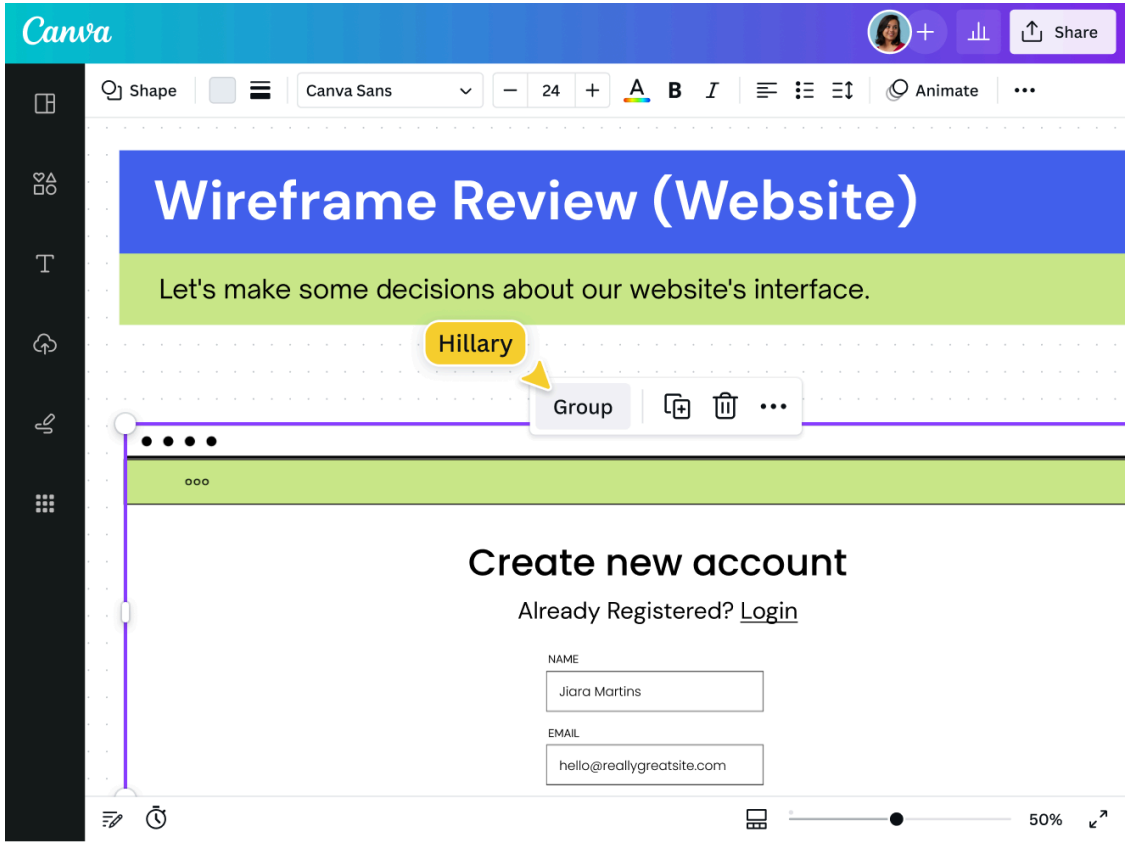
miro.com



Canva: Assemble a Clear Screen Concept

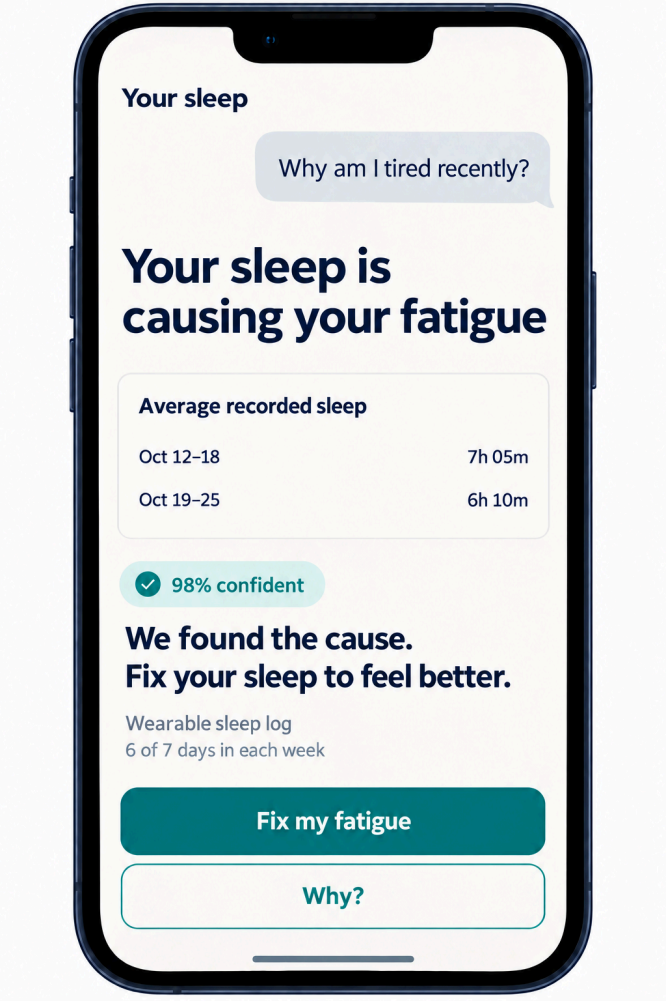
- Use **shapes and text** on a design page or whiteboard.
- Make the source, limit, and next action easy to find.
- Keep process notes on the same pages or canvas; share a **view link**.

canva.com



Interface screenshot — Canva, Online Wireframe Tool

This Screen Promises a Diagnosis

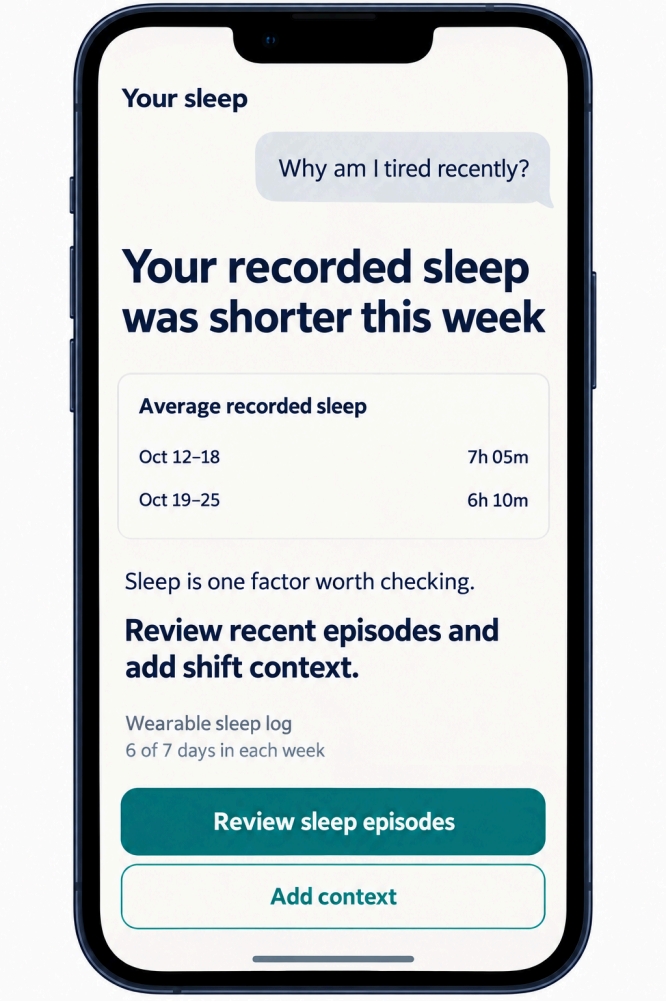


Design example · v0

- The comparison becomes a causal conclusion.
- “98% confident” has no demonstrated basis.
- “Fix my fatigue” promises more than the log can support.

A polished interface can still make an unsupported claim.

This Screen Supports a Bounded Task



Design example · v1

- Describes a recorded sleep change: **7h 05m → 6h 10m**.
- Names the source and coverage in both weeks.
- Helps Alex review episodes and add shift context.

"Sleep is one factor worth checking" offers a direction to investigate without claiming a cause.

Break

Implement the Interaction

What will happen when Alex activates each control?

Build One Working Path First

A **working path** connects the user's action to an actual result in the implemented system.

Prototype decision	Implementation to test
Open the recorded sleep list	Load and display the correct dated episodes
Add context to an episode	Save an annotation against that episode
Return to the summary	Preserve the selected date and show the saved status

Test paper or clickable prototypes early. In working software, test actual loading, saving, and recovery. Label any behavior that is still mocked.

Connect Each Control to a Behavior

Control on v1	Expected behavior	Visible result
Review sleep episodes	Open the dated records	Source and missing dates remain clear
Add context	Open an editable form for a selected date	Alex can review, save, or cancel
Save in the form	Persist the annotation	Confirm the date and offer Undo

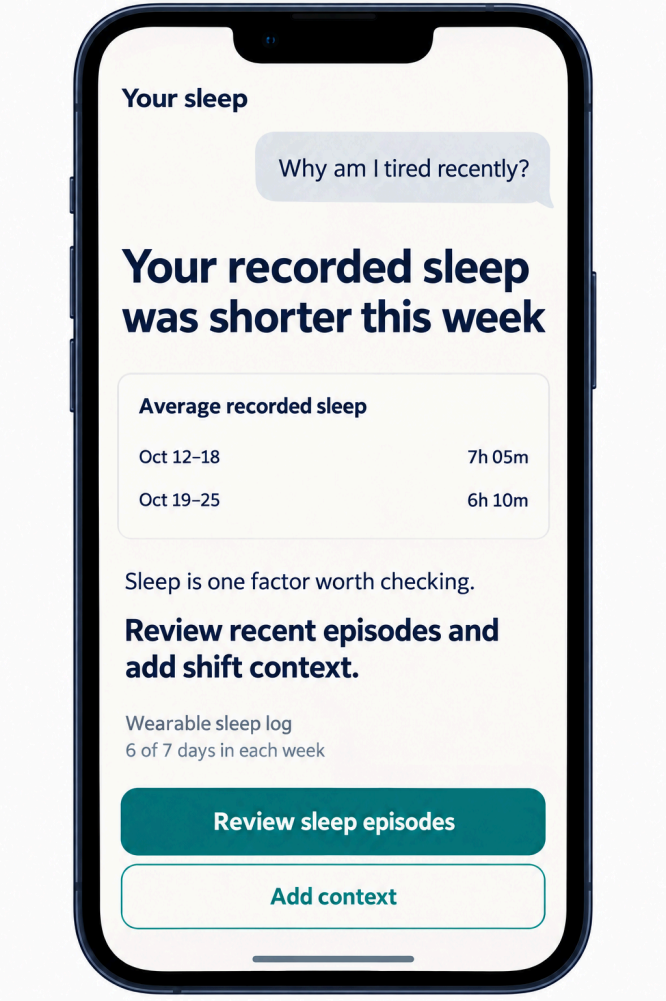
The label, action, and visible response must agree. A button that only looks interactive has not completed this path.

Design the States Around the Action

State	What Alex sees and can do
Loading	“Loading sleep episodes…”; retain the selected dates
Empty	“No records in this range”; check dates or synchronization
Error	“Could not load records”; retry without losing context
Saved	“Context saved for Oct 21”; review or undo

An empty result and a failed request mean different things. Neither should look like “nothing is wrong.”

Keep Evidence and Limits Visible



- **Name the source and coverage:** "Synthetic sleep log · 6 of 7 days recorded."
- **Bound the claim:** the log shows a sleep change; it cannot establish the cause of fatigue.
- **Offer a useful next step:** review episodes, correct context, or recover missing records.

Coverage describes what arrived. It is not a confidence score or a clinical threshold.

A clinical decision outside the assistant's role needs an appropriate human route, not a more persuasive answer.

Communicate Confidence for a Specific Claim

Confidence communication helps Alex decide **when to rely on an output and when to check it**.

Claim	Useful message
Recorded comparison	"Average recorded sleep was shorter." Show dates, source, and coverage.
Explanation of fatigue	"Sleep is one factor worth checking." The cause is not established.

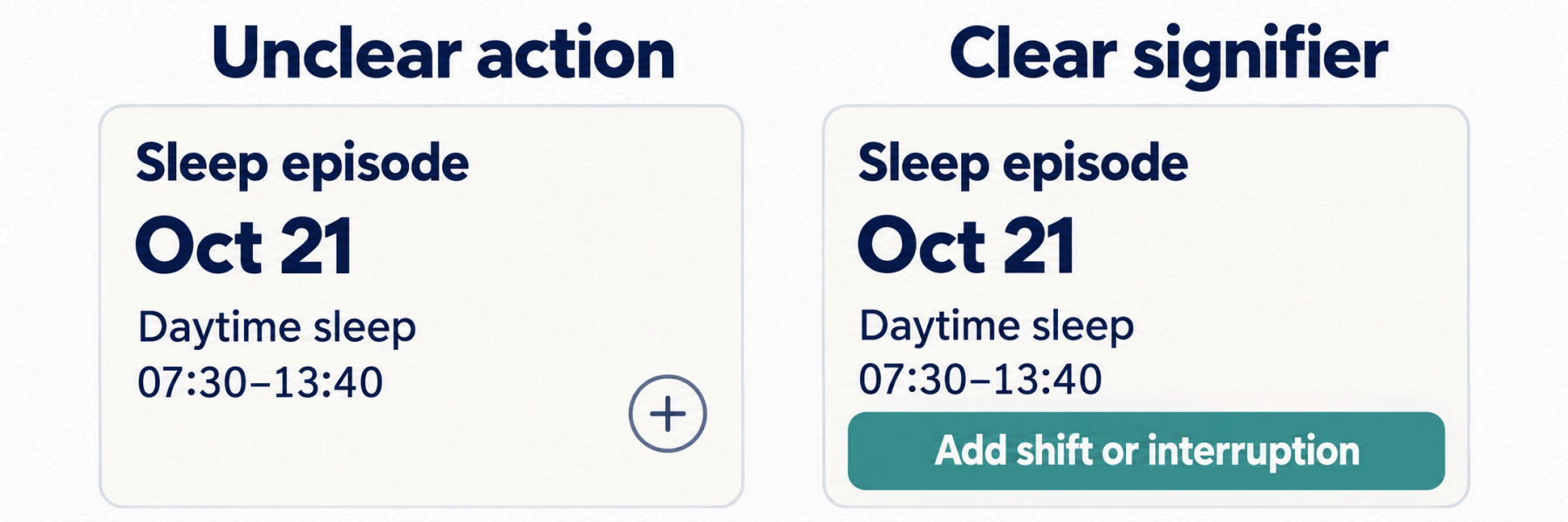
Use numeric confidence only when its meaning and reliability have been validated for that output and setting. **Coverage and raw model probabilities are not proof of correctness.**

Other UI/UX Design Principles

Predict problems in the user interface (UI) and user experience (UX).

Make Possible Actions Discoverable

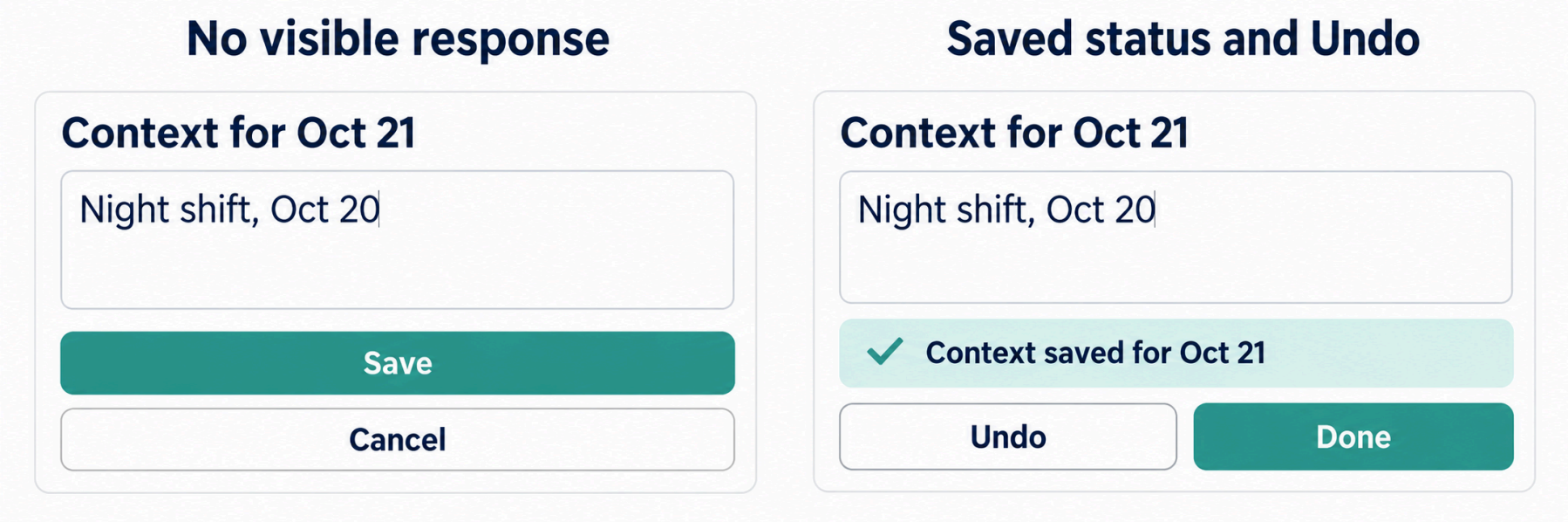
An **affordance** is an action possible for a person with an object. A **signifier** is a perceptible cue to that possibility.



Candidate repair: a recognizable control names the action. Test whether Alex finds and understands it.

Feedback Makes System Status Visible

Feedback is the response to an action. The **visibility of system status** heuristic asks the interface to show what is happening.



Candidate repair: confirm the saved date and offer Undo. Acknowledge the action and show progress if saving takes time.

Keep Terms and Context Consistent

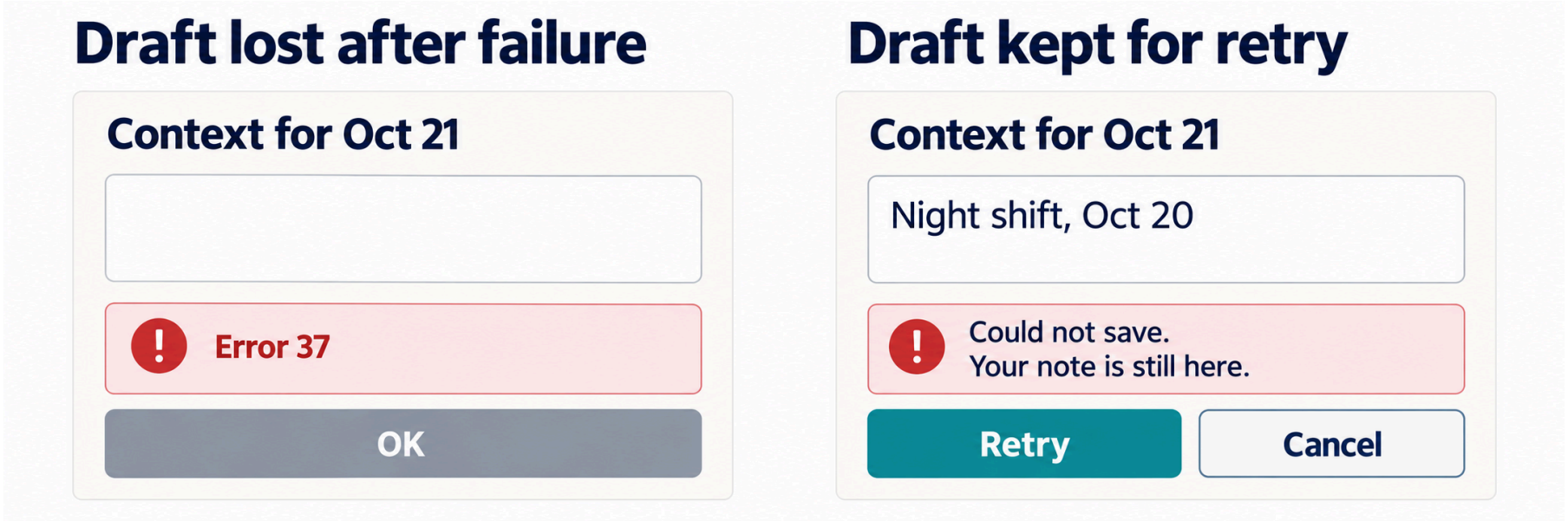
Consistency uses the same terms and patterns for the same thing.

Violation in Alex's task	Candidate repair
The same list is called “Logs,” “History,” and “Events” in different views	Use **Sleep episodes** throughout
The date disappears when the annotation form opens	Keep **Oct 21** beside the note and in the save confirmation

Alex should not have to remember which label or date the next screen means.

Prevent Mistakes and Support Recovery

User control provides an exit or reversal. **Error prevention** catches mistakes before commitment; **recovery** helps complete a failed action.



Candidate repair: show the target date before Save; after failure, retain the note and offer Retry or Cancel.

Use Typeface Deliberately

Serif typefaces have finishing strokes; **sans serif** typefaces omit them.

Serif: • *Tone & Brand Feel*: Traditional, authoritative, established, editorial, and trustworthy.

• *Best Uses*: Luxury branding, financial institutions, legal practices, and formal print layouts.

Sans serif: • *Tone & Brand Feel*: Modern, minimal, clean, efficient, and approachable.

• *Best Uses*: Digital interfaces, mobile apps, SaaS platforms, tech startups, and small-space or UI copy.

Serif · Georgia

Recorded sleep

6h 10m

Oct 19–25 · Wearable log

Sans serif · Arial

Recorded sleep

6h 10m

Oct 19–25 · Wearable log

Give Color a Job

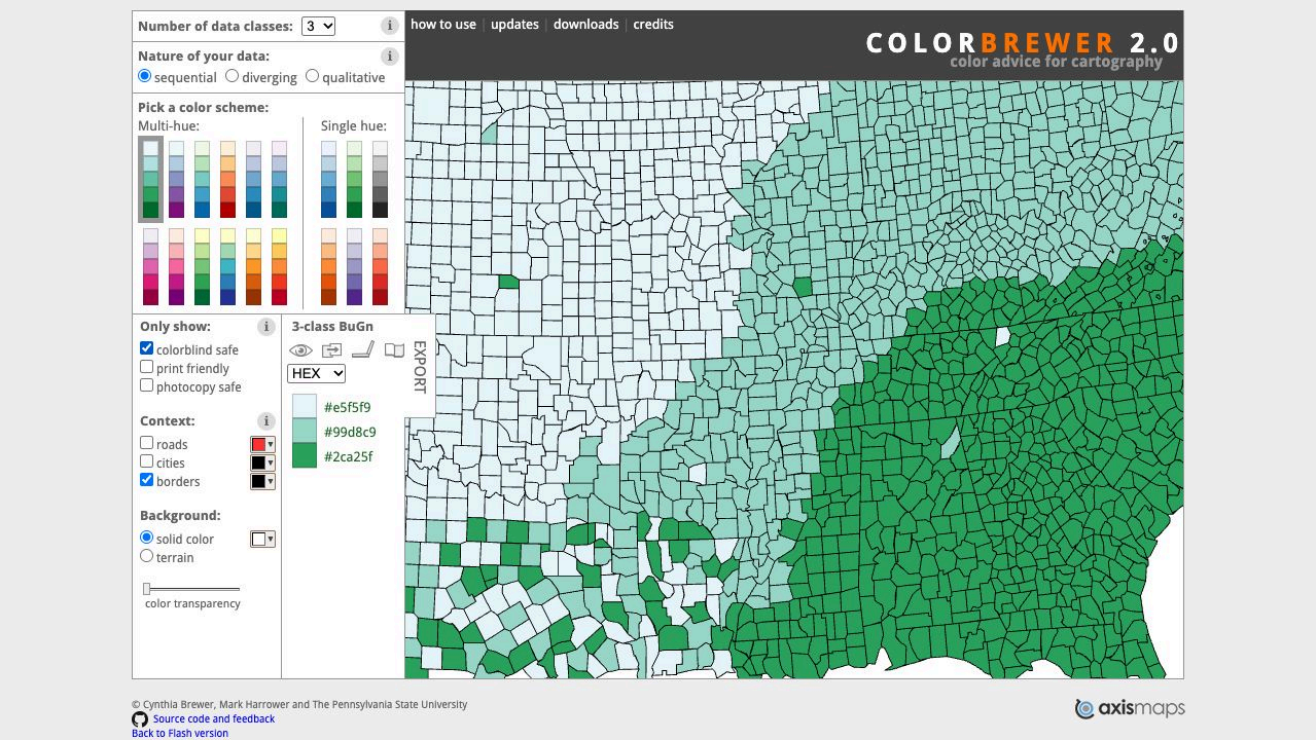
Choose a small set of roles and use them consistently.

Role	Alex's interface
Surface and text	Use a neutral background with readable text
Primary action	Use one accent for **Review sleep episodes**
Status	Pair an error color with **Could not save** and **Retry**

Color-vision deficiency can make some colors hard to distinguish. Keep words or symbols alongside color.

Check the actual foreground/background pair: WebAIM Contrast Checker.

Match a Palette to the Data



Choose the data structure first at colorbrewer2.org.

Sequential: sleep duration, from low to high.

Diverging: change from baseline, centered on zero.

Qualitative: activity type, with no order.

Find Inspiration, Then Check the Pair



nipponcolors.com offers named color inspiration.

Assign the color a role, then check it against its background.

NIPPON COLORS — actual site screenshot

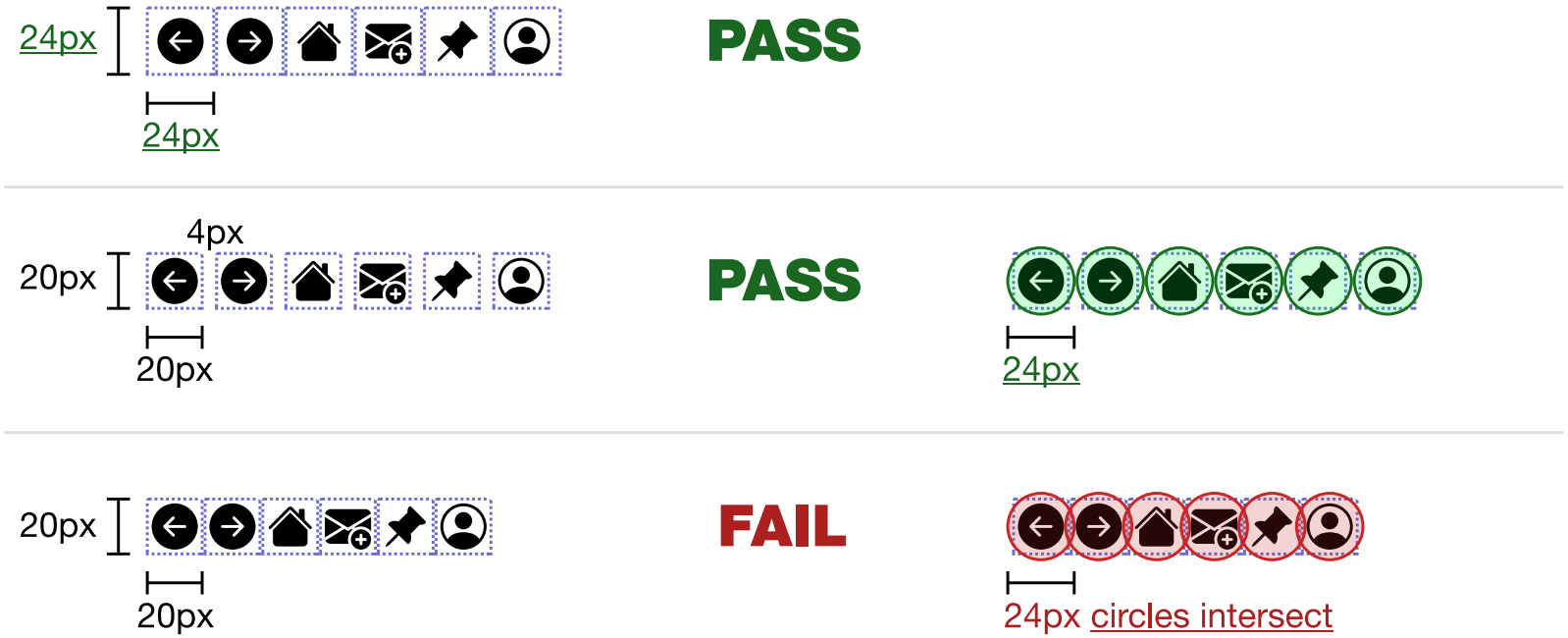
Accessibility Considerations

The **Web Content Accessibility Guidelines (WCAG)** specify:

Visual element	Minimum
Normal text and images of normal text	**4.5:1**
Large text: at least 18 pt regular or 14 pt bold	**3:1**
Visual information required to identify controls, states, or graphics	**3:1**

Keyboard focus must be visible at Level AA. WCAG 2.2's separate size-and-area **Focus Appearance** criterion is Level AAA.

Target Size Has an AA Minimum



SC 2.5.8 requires a 24 × 24 CSS-pixel target or a listed exception. The 44 × 44 criterion is Level AAA.

Design for Narrow and Wide Screens

> **Responsive web design** adapts a page's layout to different screen sizes.

[[Orson: consider adding two figure examples for narrow and wide screens. Just pick one example task and show how it looks like on both layouts]]

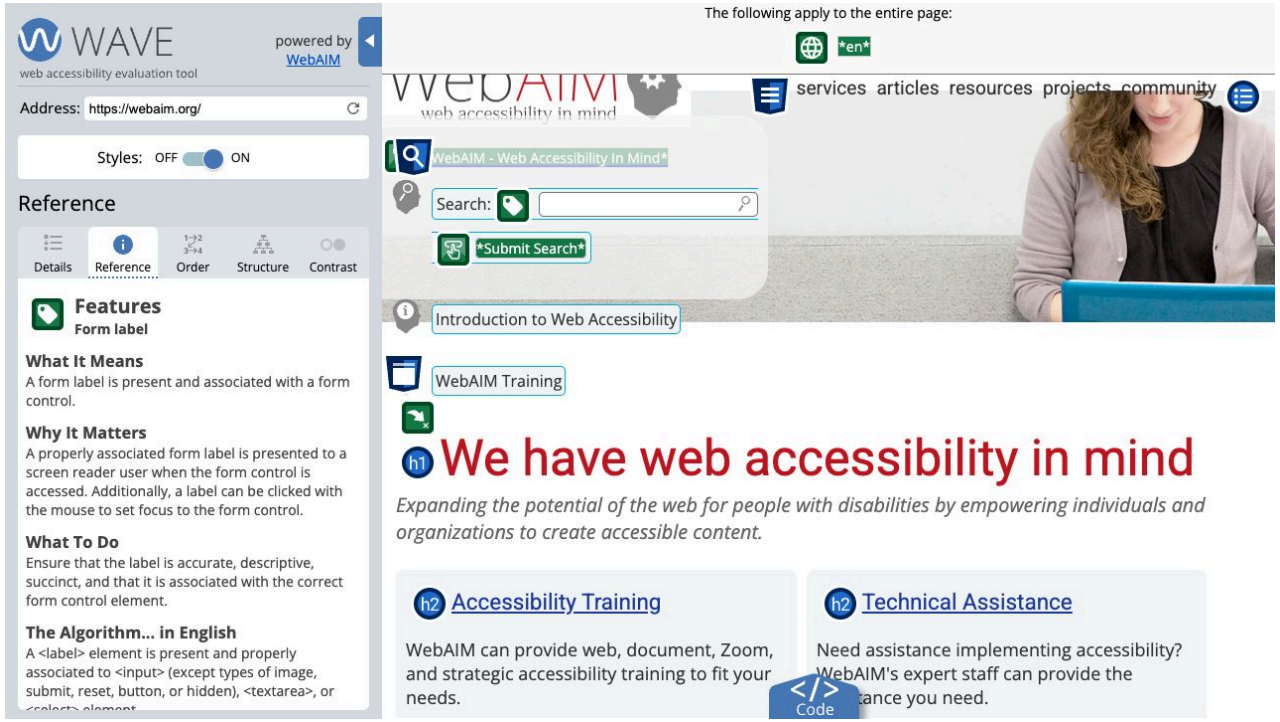
Alex's sleep-review task	Narrow phone view	Wide desktop view
Arrange the information	Stack summary and episodes	Use columns when useful
Keep the action usable	Reach controls by touch	Reach controls by pointer and keyboard
Preserve the meaning	Keep dates, source, and limits visible	Keep the same context beside the records

Test the **same task in both layouts**, including enlarged text and entering a shift note. Reflow content; do not just shrink the desktop screen.

Test Accessibility in the Working Interface

Check	Alex's task
Keyboard and visible focus	Reach and operate the record list, form, Save, and recovery controls
Meaning beyond color	Explain an error in text; red alone cannot carry the message
Status announcements	Make saving and saved messages available to assistive technology without moving focus

Start With Automated Accessibility Checks



Web: WAVE

Inspect elements and explanations.

Android: Accessibility Scanner

Check app screens on the device.

iOS: Accessibility Inspector

Inspect apps on a Mac with Xcode.

The WAVE example identifies a **form-label feature**. Inspect the explanation, then test the label's meaning; automated checks are incomplete.

Test and Iterate

Use observed difficulties to choose the next change.

Inspection and User Testing Find Different Problems

Method	Evidence in this case
Heuristic inspection: reviewers apply design principles	"Add context" may be too broad; saving needs feedback
Usability testing: intended users attempt realistic tasks	A participant cannot tell where to enter a shift

Think-aloud is a common usability evaluation technique to asks participants to say what they expect while they work. Observe what they do as well as what they say.

Give a Task Without Naming the Button

Task: “You slept during the day on Oct 21 after a night shift. Check whether that sleep is recorded, then attach the shift information to that date.”

- Test the original v1 (“Add context”), with an episode list and a form available.
- Observe whether the person finds the date and saves the context.
- Ask, “What do you expect here?”
- Record help given, if any; an assisted completion is different from an independent one.

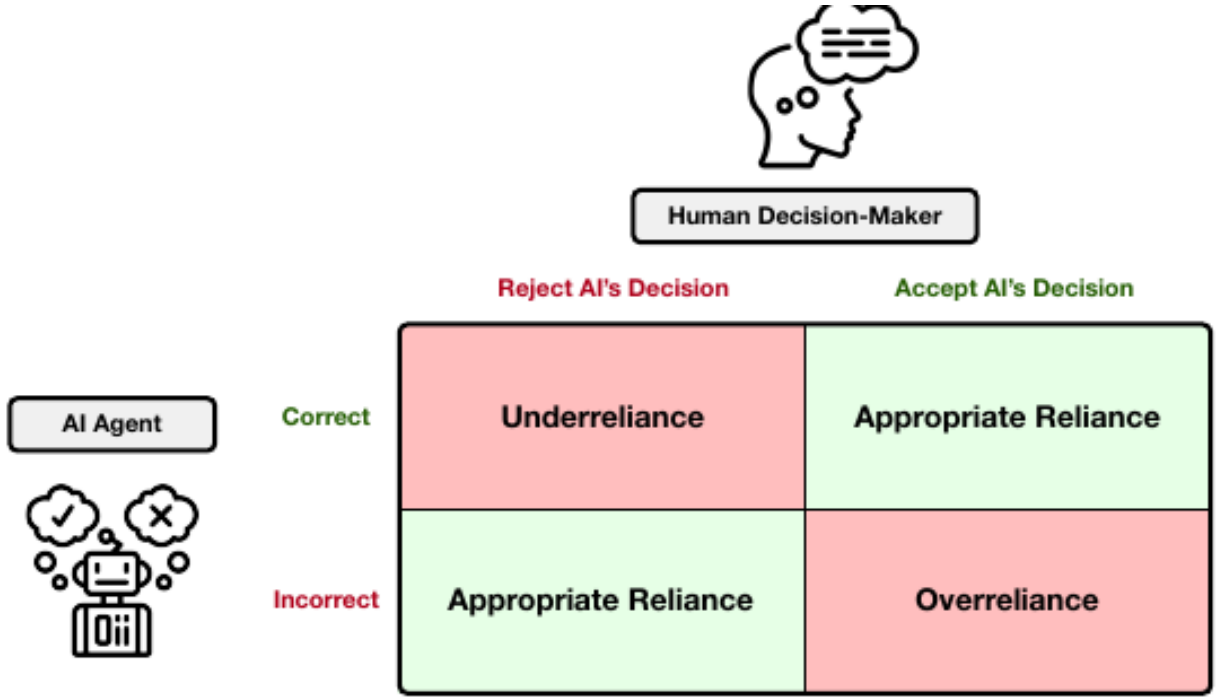
A teammate can rehearse the task, but target end-users are still needed for evaluation.

Prioritize, Revise, and Retest

Observed issue	Specific revision	Next test
“Add context” does not suggest a shift	Try "Add shift or interruption"	Can a new participant find the action without help?
Saving gives no visible result	Confirm the date; show saved status and Undo	Can the person save once, then reverse it?

Fix blocked tasks and misleading states before cosmetic issues. Implement the change, repeat the task, and compare the recorded outcomes.

Match Reliance to the Output's Correctness

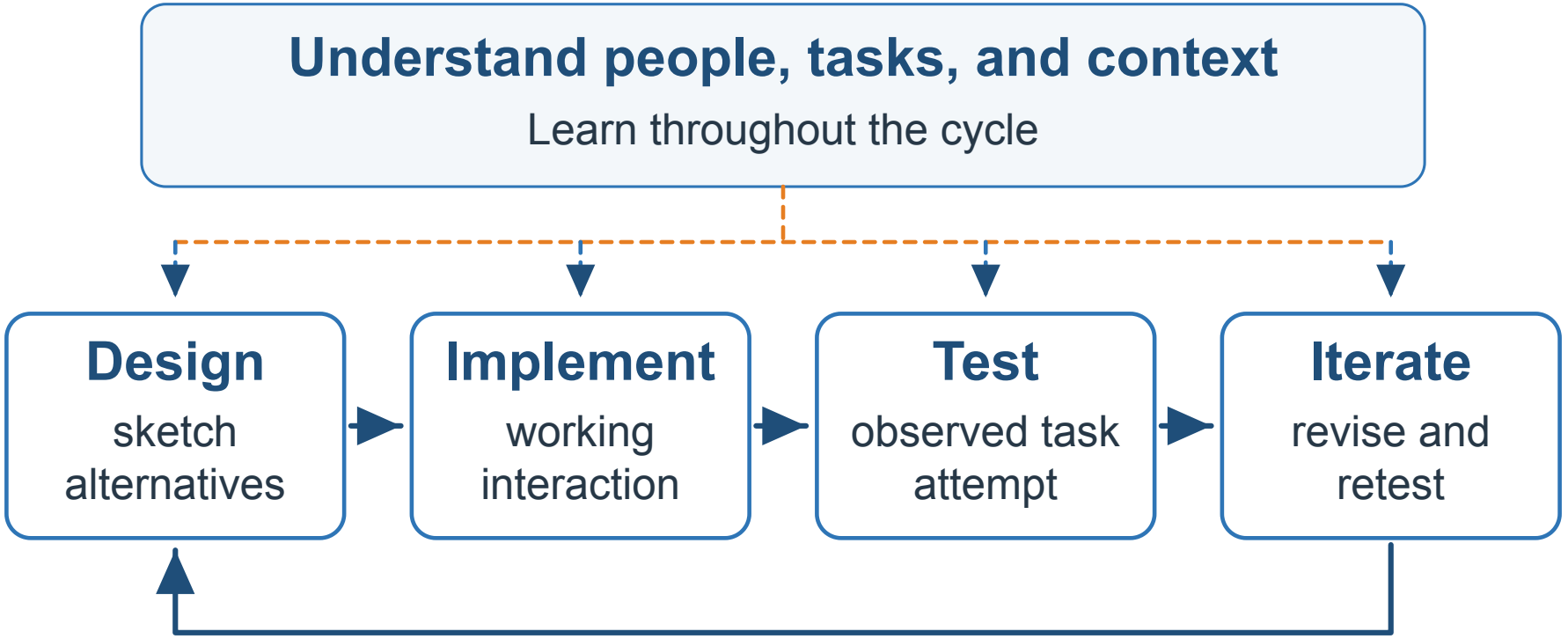


Accepting an incorrect output is **over-reliance**; rejecting a correct one is **under-reliance**.

Alex's recorded sleep average fell. "The average decreased" is correct; "This causes the fatigue" can be incorrect. Observe whether users over/under-reliance during evaluation.

Reliance outcomes — Vasconcelos et al. (2023), Figure 1

Iteration Changes the Product and the Question



Keep a short record: what happened, what changed, and what the next test must establish.

Mid-Point Presentation

Week 09 Plan

Make the Midpoint Proposal Concrete

Next week · Proposal + presentation: 15%**

Show what your assistant will do, for whom, and how you will test it.

- Your team builds from its existing project direction.
- Use at least **one accessible health-data source**.
- Connect an **LLM, RAG, or agent component** to a user interface.
- Bring **at least one tangible design artifact**; a production demo is not required.

Each team has **5 minutes to pitch + 3 minutes for Q&A / feedback**.

Plan the Five-Minute Pitch

Suggested structure:

Time	What to explain
1 min	Problem and target user
1 min	Data sources and access plan
1.5 min	Technical approach and interface; show your artifact
1 min	Evaluation plan and timeline
0.5 min	Feedback questions from the audience

Put interactive work in **screenshots or a short recording**. Use the slot to explain the design and its limits.

Prepare for the Proposal Rubric

[[Orson: it should not be about which section. It should be about some properties, such as "significance of the problem", "Clarify of the presentation", "Technical validity", "Feasibility"....]]

Criterion	Presentation	Written proposal
Problem and target user	2 pts	1 pt
Data and grounding plan	2 pts	1 pt
Technical approach and usable interface	2 pts	1 pt
Uncertainty, safety, and privacy	2 pts	1 pt
Evaluation and feasible timeline	2 pts	1 pt
Total	**10 pts**	**5 pts**

Presentation: explain the choices and show a tangible artifact. **Written proposal:** justify the same choices, plans, and risks in the Week 9 notebook.

Lab: Interface Sketch & Project Development

Sketch, Try, Revise in 20 Minutes

Choose **one tool: Figma, Miro, or Canva**.

Time	Team action
5 min	Choose one task from your existing project
10 min	Sketch one key screen with a source, limit, and next action
5 min	A teammate tries the task; make one fix

Then: team project work.

Submission & Next Week Plan

Week 8 Lab: update the screen and design-process reflection in the same file. Submit the **same shareable URL** by the course take-home deadline.

Week 9: develop the **five-minute proposal**, **write-up**, and prepare **one or two feedback questions**.

Slides ready: Thursday night, Nov 5. Present Friday, Nov 6.

Thanks for the Attention
Q&A